Gynaecomastia – Aetiology & Management

Gynaecomastia

- benign proliferation of the glandular tissue of the male breast
- caused by an increase in the ratio of estrogen to androgen activity
- may be unilateral or bilateral
- diagnosed on exam as a palpable mass of tissue at least 0.5 cm in diameter (usually underlying the nipple).
- common in infancy, adolescence, and in middle-aged to older adult males.

Important to differentiate between

1) Pseudogynaecomastia, which is often seen in obese men, refers to fat deposition without glandular proliferation, and does not require evaluation.
2) Breast carcinoma, which is far less common.

Causes

Common

Physiologic	**Pathologic**
Neonatal Pubertal Involutional } 25%	Drugs (25%) Idiopathic (25%) Cirrhosis or malnutrition (8%) Male hypogonadism – primary (8%) or secondary (2%) Neoplasms (3%) - Testicular – germ cell, Leydig, Sertoli, sex cord - Adrenal – adeno/carcinoma - Ectopic production of hCG Hyperthyroidism (1.5%) Renal disease & dialysis (1%)

Rare

Enzymatic defects in testosterone production Androgen-insensitivity syndromes True hermaphroditism Excessive extraglandular aromatase activity

Puberty - no differences in single-point measurements of serum concentrations of testosterone, estradiol, estrone, or gonadotropins. However, some reports have shown a transient increase in estradiol concentration at the onset of puberty in boys who develop gynecomastia. These boys may also have wider fluctuations of estradiol levels, with an absolute increase in 24-hour concentration of estradiol, which may reflect increased conversion of adrenal androgens to estrogens.

During puberty, the serum estradiol concentrations rise to adult levels before the testosterone concentration. It is likely that this transient imbalance accounts for much of the estrogen/androgen imbalance that leads to pubertal gynecomastia.

Pubertal gynecomastia usually resolves spontaneously within six months to two years of onset, but in some instances, may persist after completion of puberty into adulthood, resulting in persistent pubertal gynecomastia.

Drugs — Very common cause. Some are predictable (ie anti-androgen therapy), whilst others are less expected (ie Spironolactone, cimetidine)

Spironolactone can increase the aromatization of testosterone to estradiol, decrease the testosterone production rate by the testes, and displace testosterone from SHBG, thereby increasing its metabolic clearance rate. Spironolactone also can act as an antiandrogen by binding to androgen receptors and displacing or preventing binding of testosterone and dihydrotestosterone to their receptors. Studies have shown that low dose spironolactone has 10% chance of causing gynaecomastia, and almost all of men who take high dose spironolactone.

**Even drugs within the same class do not all cause gynecomastia to the same extent.

CCB - nifedipine has the highest frequency of gynecomastia and diltiazem the lowest. Thus, in an older man who is at increased risk of gynecomastia simply on the basis of age, diltiazem would be preferable to nifedipine.

H2-receptor/PPI - the incidence of gynecomastia is highest with cimetidine and then ranitidine, and is lowest with omeprazole. Thus, omeprazole would be a better choice in an older individual with other risk factors for developing gynecomastia.

Gynecomastia occurs in over 50 percent of men with prostate cancer receiving estrogen or antiandrogen monotherapy.

Herbal products - Tea tree oil and lavender oil

HAART — men with HIV on highly active antiretroviral therapy is usually due to fat tissue (lipomastia or pseudogynecomastia) as part of a fat redistribution syndrome (lipodystrophy) esp efavirenz

Idiopathic — Gynecomastia in adults is often multifactorial. Aging per se is associated with an increase in body fat relative to the lean body mass. Adipose tissue is an active site of extraglandular aromatization of testosterone to estradiol and of androstenedione to estrone. In addition, there tends to be a gradual decrease in testosterone production by the aging testes and an increase in SHBG levels, resulting in a fall in the free testosterone concentration with a reciprocal increase in the luteinizing hormone (LH) level. These two factors probably account for most patients who have "idiopathic" gynecomastia. Older men are also more likely to take medications associated with gynecomastia then are younger men.

Cirrhosis or malnutrition — 67% of cirrhotic pts have gynaecomastia. Studies inconclusive as to whether this is due to age cohort, drugs (ie spironolactone) or actually due to cirrhosis

During starvation, both gonadotropin and testosterone levels were probably reduced, while estrogen production was probably normal due to normal estrogen production from adrenal precursors. These changes will promote the development of gynecomastia. During refeeding, gonadotropins rise, resulting in both an increase in testosterone secretion and a marked elevation in estradiol production that mimics normal puberty. Thus, patients who develop refeeding gynecomastia may be said to have undergone a "second puberty".

Male hypogonadism — Primary hypogonadism can be due to a congenital abnormality such as Klinefelter's syndrome or an enzymatic defect in the testosterone biosynthetic pathway, or to testicular trauma, infection, infiltrative disorders, vascular insufficiency or aging.

Secondary hypogonadism due to a hypothalamic or pituitary abnormality may also be associated with gynecomastia. In these patients, the production of LH is deficient, resulting in a low testosterone production rate and low estradiol production from the testes. However, the adrenal cortex continues to produce estrogen precursors which are aromatized in extraglandular tissue; the result is an estrogen/androgen imbalance.

Testicular neoplasms — Germ cell tumors account for approximately 95 percent of testicular neoplasms; between 2.5 percent and 6 percent of affected patients have gynecomastia at the time of presentation. The gynecomastia is associated with secretion of hCG by foci of choriocarcinoma or trophoblastic cells in the tumor. Poor prognostic sign if present at the time of diagnosis of the tumor.

Also occurs in 15 percent of patients after successful treatment with surgery, chemotherapy, or radiation therapy; at this time, it does not affect predicted survival. This form of gynecomastia results from hypogonadism secondary to the chemotherapy or radiation; hCG is not found in the serum. It often spontaneously resolves within one year

Hyperthyroidism — Gynecomastia has been reported in 10 to 40 percent of men with hyperthyroidism due to Graves' disease. Serum LH levels are often elevated, contributing to increased estradiol relative to testosterone production by Leydig cells. There is also enhanced aromatization of testosterone to estradiol and of androstenedione to estrone in extraglandular tissues. This results in increased concentration of SHBG. Free testosterone levels are normal or low, while free estradiol levels are elevated. Thus, gynecomastia results from the combination of decreased free androgen levels combined with the overproduction of estrogens.

Chronic renal failure — occurs in abt 50% of patients treated with maintenance hemodialysis. The primary cause of the gynecomastia appears to be Leydig cell dysfunction. Serum testosterone levels are low and gonadotropins are appropriately elevated; the metabolic clearance of LH is also reduced. Gynecomastia may occur following renal transplantation as gonadal function improves ("refeeding gynecomastia") and/or the use of transplantation medications such as cyclosporine.

Other rare causes

Feminizing adrenocortical tumors — These are rare tumors which are malignant in approximately 75 percent of cases, with a median survival of 1.5 years. The combination of increased secretion of estrogens by the tumor and increased peripheral aromatization of adrenal androgens to estrogens accounts for the gynecomastia.

Ectopic hCG — Precocious puberty can occur in boys with hCG-secreting hepatoblastomas. In adults, large cell carcinoma of the lung, gastric carcinoma, renal cell carcinoma and occasionally hepatoma have been associated with gynecomastia and marked elevations of serum hCG.

True hermaphroditism — True hermaphrodites harbor both testicular and ovarian tissue, and may develop gynecomastia from excessive estrogen secretion by the ovarian component.

Androgen insensitivity syndromes — The androgen insensitivity syndromes are a group of disorders due to defects in or absence of the intracellular androgen receptor in androgen target tissues

Aromatase excess syndrome — Familial prepubertal gynecomastia is a rare disorder of increased aromatase activity resulting in severe estrogen excess. autosomal dominant.

It is possible that some patients with the diagnosis of "idiopathic" gynecomastia actually represent excessive extraglandular aromatase activity.

Management

GENERAL PRINCIPLES

The management of gynecomastia depends upon its etiology, duration, severity, and presence or absence of tenderness. In many cases, gynecomastia resolves without therapy.

1) Etiology
    a. Spontaneous regression — Gynecomastia of recent onset (less than 6 months) in both adolescents and adults will often regress spontaneously. Thus, observation alone is a reasonable approach in most patients.
    b. Pubertal gynecomastia — Pubertal gynecomastia, a benign, physiologic process, is common in adolescent boys, often developing between ages 10 and 12 years, with a peak prevalence of 65 percent between ages 13 and 14, followed by regression in 85 to 90 percent within six months to two years. Its persistence beyond age 17 is uncommon.
    c. Adults — Regression of gynecomastia is also common in adults.

2) Severity
    a. Pain is usually not severe. Varying degrees of tenderness and nipple sensitivity with rubbing against a shirt are more common than pain, and if severe, these symptoms may be indications for earlier therapeutic intervention, even though these symptoms are usually self-limited as fibrotic changes occur after 6 to 12 months.
    b. Psychological

3) Duration— major factor that should influence the initial approach
    a. First 6 months - ductal hyperplasia and periductal inflammation. During this the time period, gynecomastia is the most symptomatic and also most treatable.
    b. More than 12 months - fibrosis and disappearance of the inflammatory reaction. It is unlikely that any medical therapy will result in significant regression in the late fibrotic stage. As a result, medical therapies, if used, should be tried early in the course.

4) Avoidance of offending drugs — simplest method to prevent or manage gynecomastia is to avoid or discontinue those drugs that cause it. Remember the variations between classes.

Treatment

1) Ensure that the pt has gynaecomastia (Hx & exam) (Diagram 1)

2) Observe first – bloods as appropriate – HCG, LH, testosterone, TFTs, LFTs, renal function (Diagram 2)

3) Pharmacological

   - Androgens – Testosterone for hypogonadal men
   - SERM (tamoxifen/Raloxifene) – good for symptoms, but does not cause regression, used for severely painful breasts/ cosmesis, must be less than 12 months onset
   - ?Aromatase inhibitors – not shown to be of benefit

4) Surgical

   - After 12 months of symptoms (ie fibrosis)
   - Not recommended in children until after full gonadal development
   - Aim – restore normal breast contour, correct deformity of breast, nipple, areola
   - Options – subcutaneous mastectomy, liposuction-assisted mastectomy

5) Special group – Prostate cancer pts

   - Common with androgen blockade therapy
     - 75% if on monotherapy
     - 15% if on GnRH analogue and androgen blockade
   - Tamoxifen recommended if symptomatic
   - Alternative is RTX – prophylactic RTX has been shown to have benefit in decreasing incidence of gynaecomastia
   - Surgery if Sx more than 12 months

Diagram 1

The patient lies flat on his back with his hands clasped beneath his head. Using the separated thumb and forefinger, the examiner slowly brings the fingers together from either side of the breast. In patients with true gynecomastia, a rubbery or firm mound of tissue that is concentric with the nipple–areolar complex is felt, whereas in patients with pseudogynecomastia, no such disk of tissue is found.

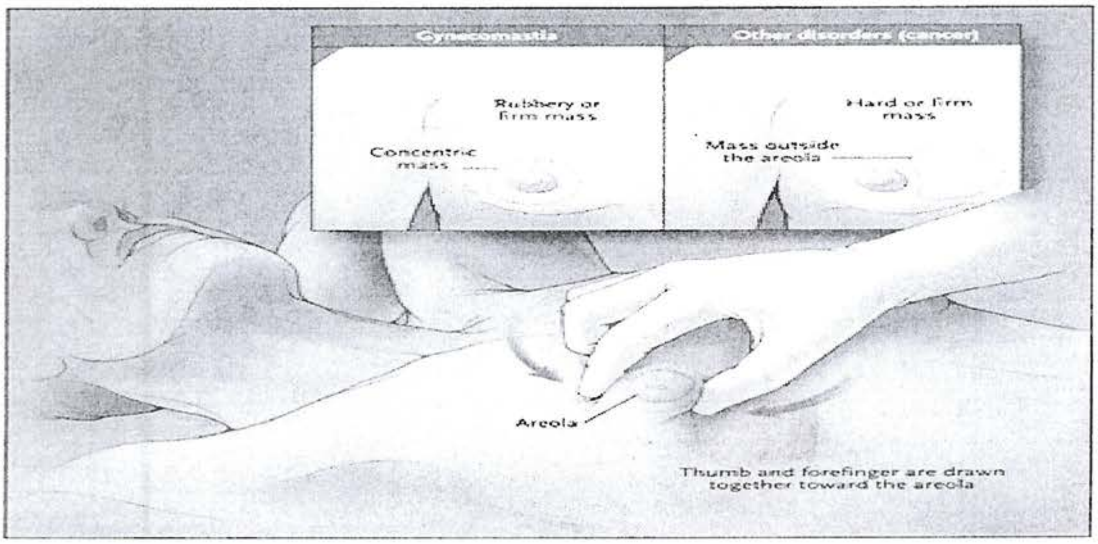

Diagram 2

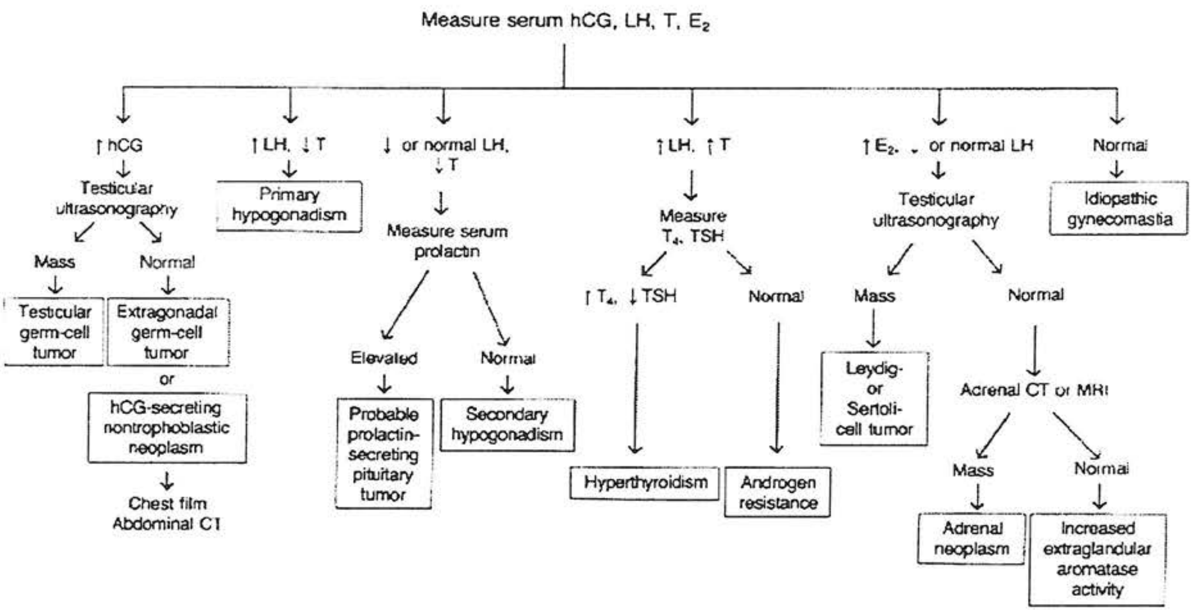

Diagram 3

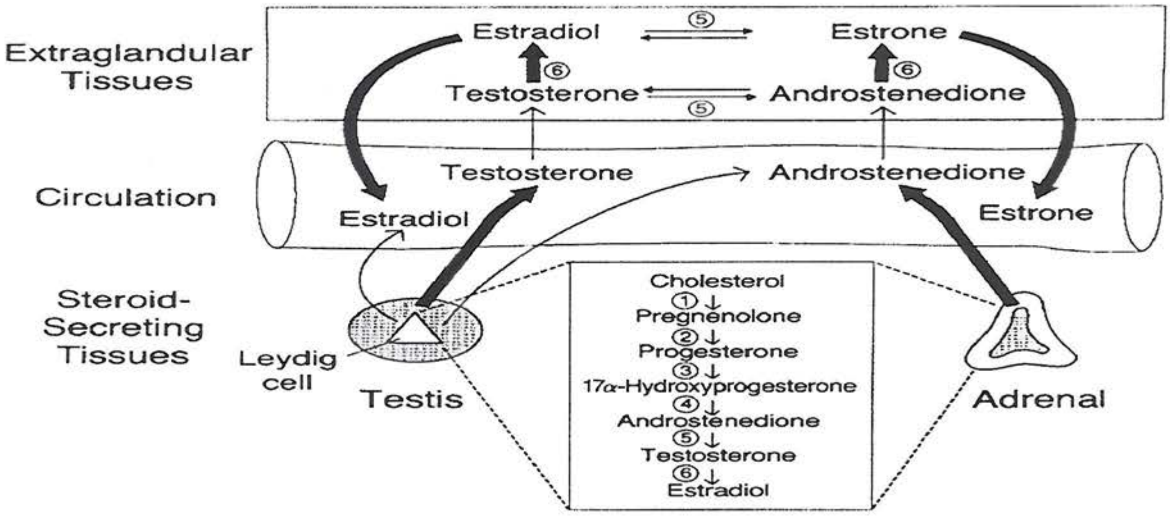

Diagram 4

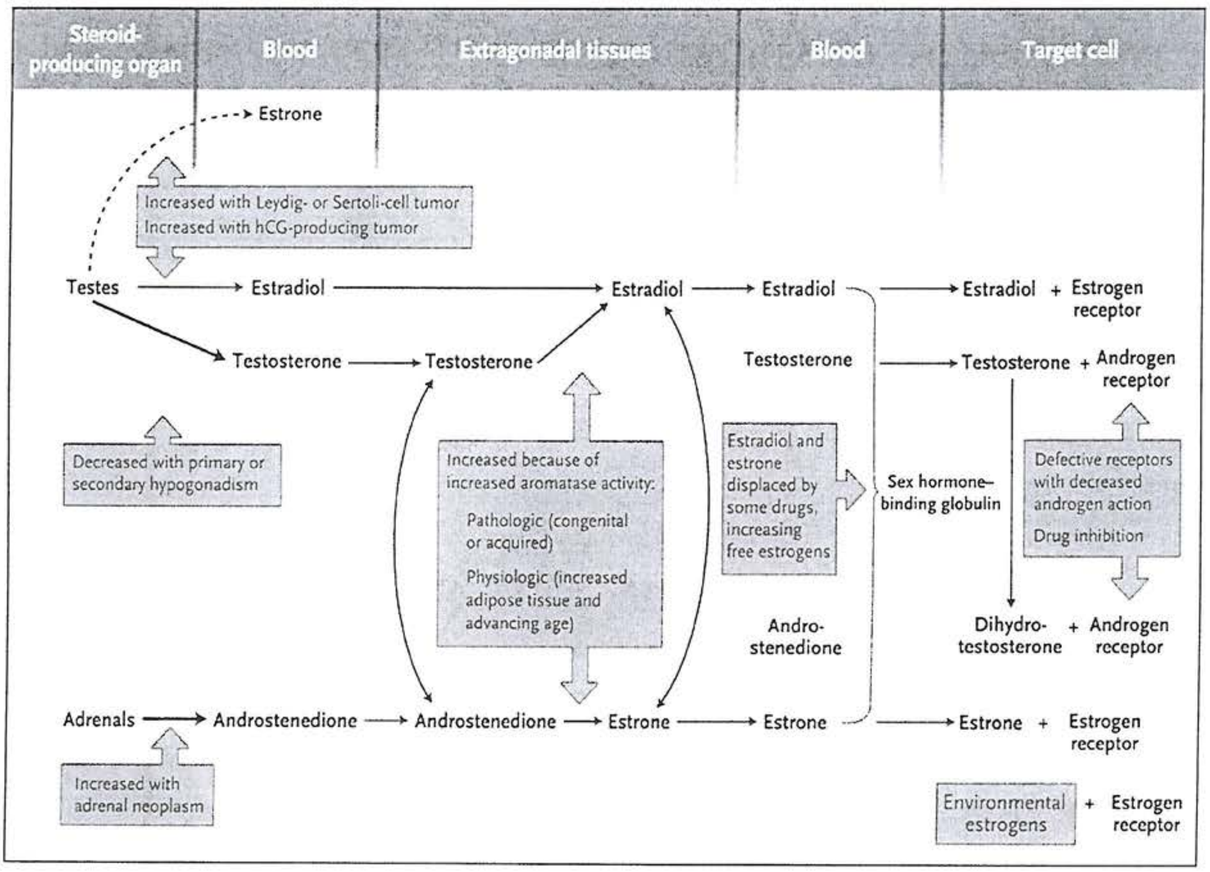

Diagram 5

Table 1. **Signs and Symptoms of Pathologic Processes That Cause Gynecomastia.**

Condition	Symptoms	Signs
Tumor		
Testicular		
Leydig-cell or Sertoli-cell	Testicular pain, enlargement, or both; decreased libido	Testicular mass or enlargement, contralateral testis with some atrophy, signs of feminization
Germ-cell	Testicular pain, enlargement, or both; symptoms of metastases (e.g., back pain, hemoptysis)	Testicular mass
Adrenocortical	Weight loss, decreased libido, possible symptoms of coexisting Cushing's syndrome or mineralocorticoid excess	Abdominal mass, signs of Cushing's syndrome or mineralocorticoid excess (hypertension)
Ectopic hCG-secreting	Weight loss; respiratory symptoms with lung carcinoma; abdominal symptoms with hepatocellular, gastric, or renal-cell carcinoma	Dependent on location of primary tumor and presence or absence of metastases
Hypogonadism		
Primary	Decreased libido, erectile dysfunction, vasomotor symptoms	Decreased testicular size, hard texture with Klinefelter's syndrome, soft texture if acquired; incomplete development of secondary sexual characteristics with prepubertal onset; possible findings of a systemic disorder (e.g., hemochromatosis)
Secondary	Decreased libido, erectile dysfunction, symptoms of other pituitary hormone deficiency, headache, visual symptoms	Decreased testicular size; possible visual-field cuts from a pituitary or parasellar tumor; signs of hypothyroidism, excess or deficiency of growth hormone; galactorrhea (rare)
Hyperthyroidism	Weight loss, palpitations, increased sweating, increased frequency of defecation, nervousness, insomnia, heat intolerance	Goiter, tremor, tachycardia, upper-eyelid retraction
Liver disease	Anorexia, nausea, vomiting, weight loss (or weight gain with ascites), edema, jaundice, pruritus	Jaundice, enlarged or shrunken liver, ascites, edema
Renal disease	Anorexia, fatigue, nausea, vomiting, oliguria or polyuria, pruritus, yellowish skin	Lethargy, asterixis, uremic hue, hypertension
Androgen insensitivity	Decreased libido, infertility	Possible hypospadias or ambiguity of genitalia, possible neurologic findings (e.g., proximal muscle weakness with fasciculations and tremor in X-linked spinal and bulbar muscular atrophy)
Familial or sporadic aromatase excess syndrome	None	Prepubertal onset of gynecomastia; accelerated increase in height in childhood, reduced final height; incomplete virilization

Antiandrogens/inhibitors of androgen synthesis	Drugs of abuse
Cyproterone acetate	Alcohol
Flutamide, bicalutamide, nilutamide	Amphetamines
Finasteride, dutasteride	Heroin
Spironolactone	Marijuana
Ketoconazole	Methadone
PC-SPECS (OTC herbal)	**Hormones**
HAART therapy	Androgens
Antibiotics	Anabolic steroids
Ethionamide	Chorionic gonadotropin
Isoniazid	Estrogens
Ketoconazole	Growth hormone
Metronidazole	**Psychoactive drugs**
Antiulcer drugs	Diazepam
Cimetidine	Haloperidol
Ranitidine	Phenothiazines
Omprazole	Tricyclic antidepressants
Cancer chemotherapeutic drugs	**Other**
Alkylating agents	Auranofin
Methotrexate	Diethylproprion
Vinca alkaloids	Domperidone
Combination chemotherapy	Etretinate
Imatinib	Metoclopramide
Cardiovascular drugs	Phenytoin
ACE inhibitors: captopril, enalapril	Penicillamine
Amiodarone	Sulindac
Calcium channel blockers: diltiazem, nifedipine	Theophylline
Digitoxin	
Methyldopa	
Reserpine	